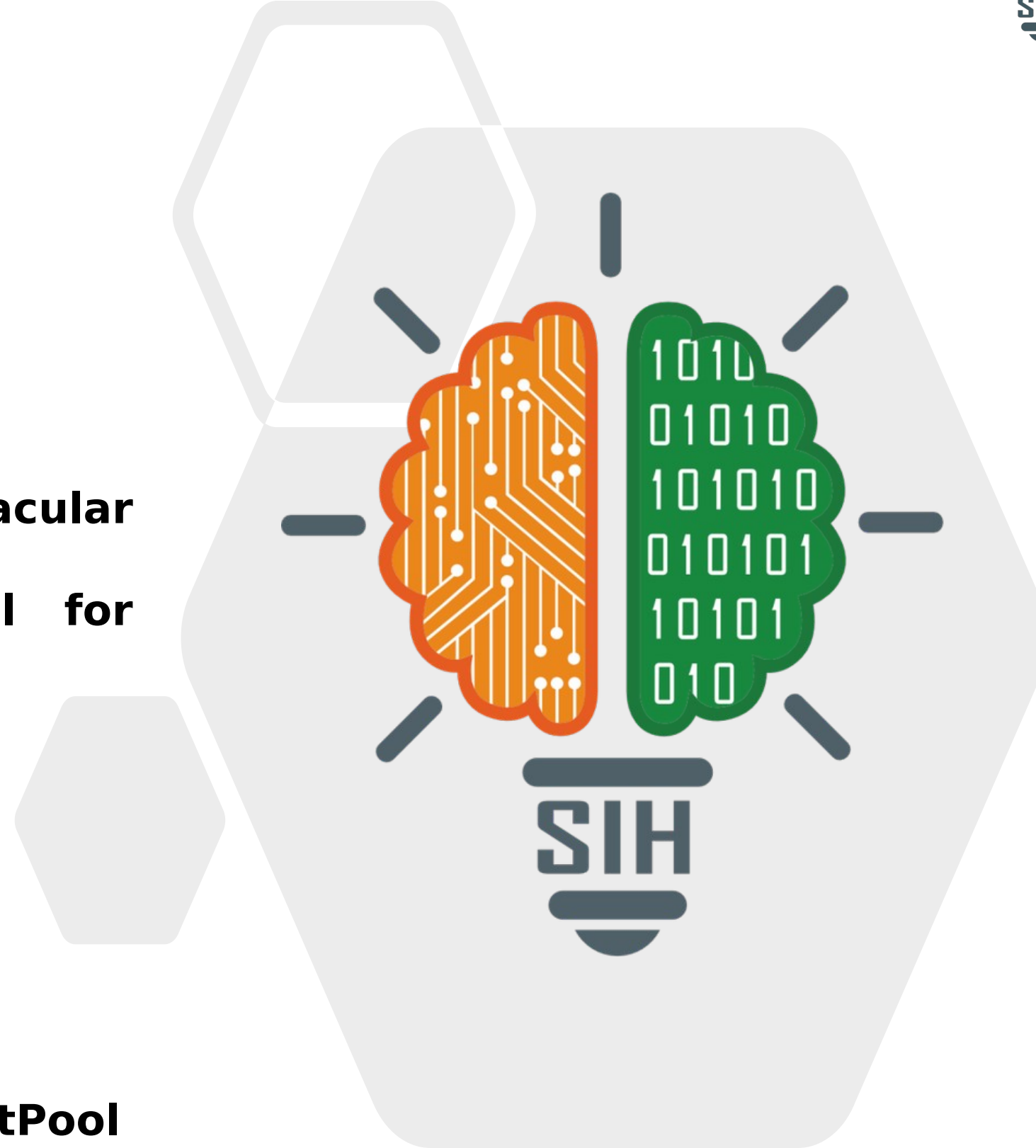


SMART INDIA HACKATHON 2026

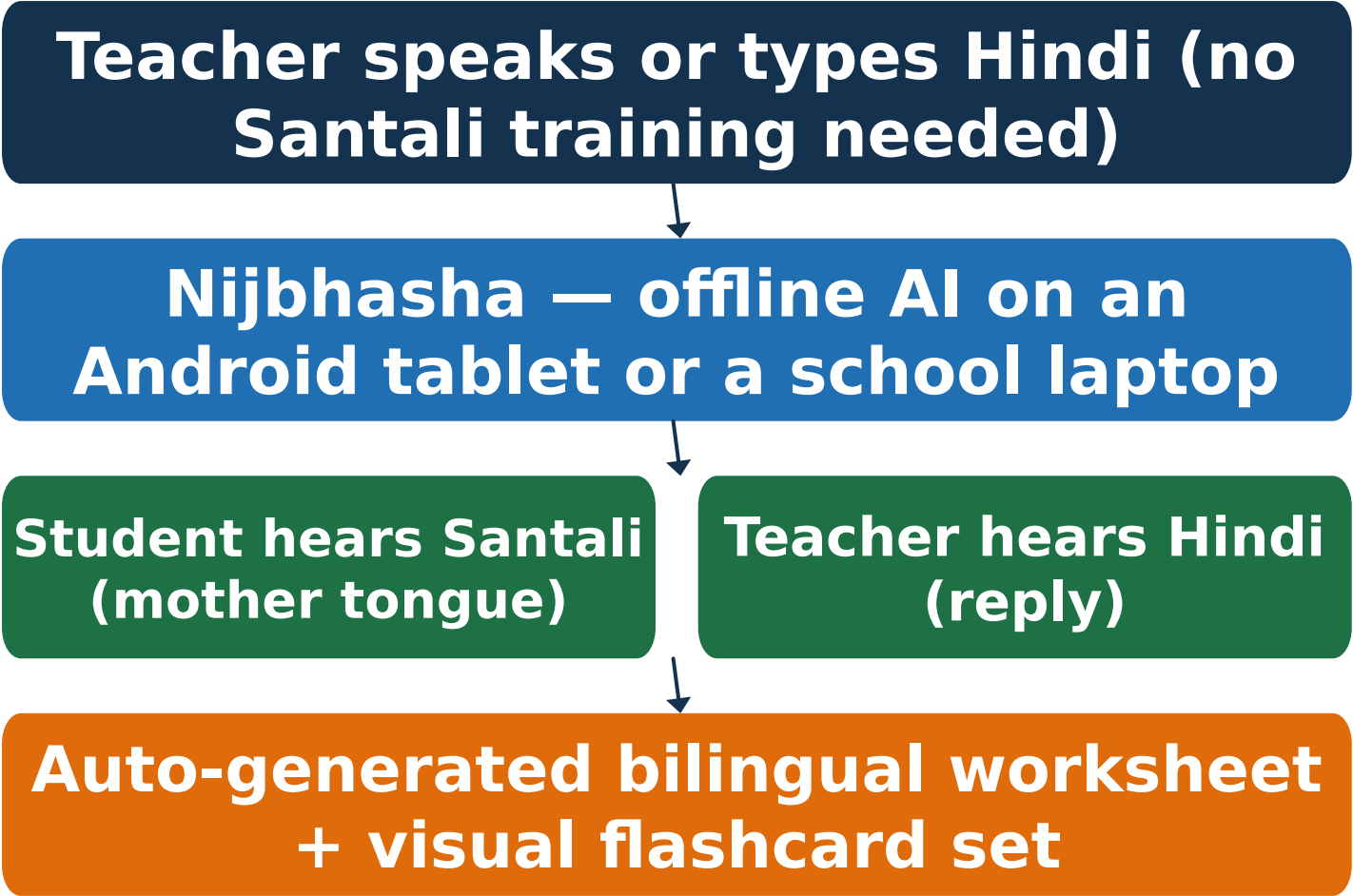
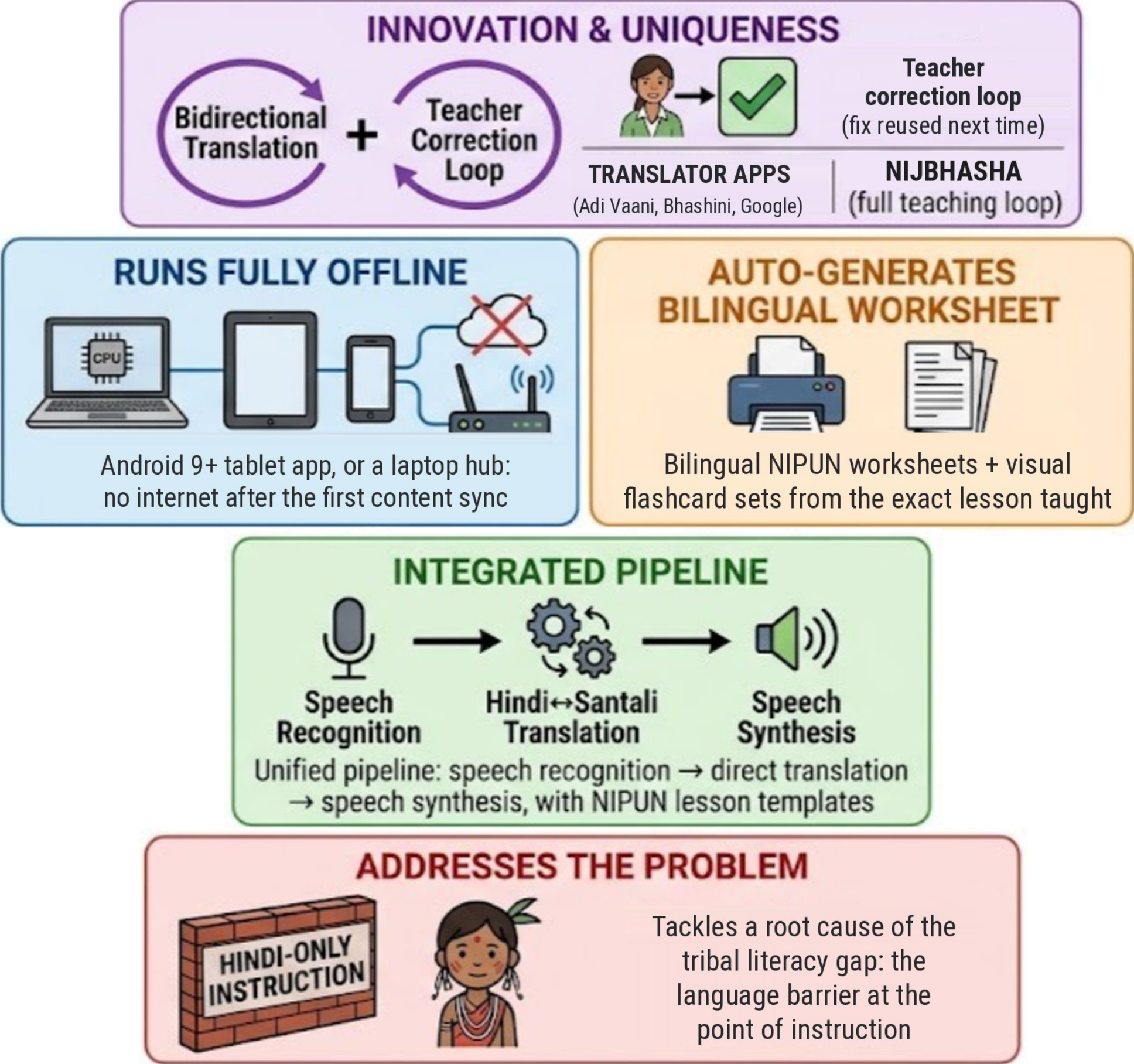


SMART INDIA
HACKATHON
2026

- **Problem Statement ID - SIH26042**
- **Problem Statement Title- AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education**
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID- 168531**
- **Team Name (Registered on portal) - VITV 8-bitPool**



Nijbhasha — Proposed Solution



- What Nijbhasha adds over translation apps**
- Adi Vaani (MoTA), Bhashini and Google Translate translate text. Nijbhasha runs the whole lesson:
- ✓ Lesson script → speech → comprehension check
 - ✓ Bilingual NIPUN worksheet + flashcards from that lesson
 - ✓ Never guesses: unknown lines refused, doubtful output flagged
 - ✓ Offline on an Android tablet in airplane mode
 - ✓ Reading-fluency check against NIPUN words-per-minute goals

• WHAT'S BUILT & RUNNING TODAY

- Android app: lessons, worksheets, spoken lesson lines, typed translation; tested offline on a 4 GB tablet and a 2 GB Android 9 emulator
- Laptop hub: Flask API + one HTML page; tablets join over Wi-Fi
- Real models live: speech recognition, translation, synthesis
- Hindi / Santali / English UI; 18 NIPUN lessons, Balvatika–Grade 3
- Teachers add lessons from Hindi text; progress by NIPUN Lakshya

• THE AI PIPELINE — BUILT, NOT PLANNED

- ASR: IndicConformer — 600M on the hub, 120M int8 on the tablet
- NMT: IndicTrans2 320M, direct Hindi ↔ Santali, no English pivot
- TTS: Piper voice reads Santali via our Ol Chiki transliterator
- Offline: sockets-off laptop test; tablet tested in airplane mode
- Santali live; Mundari translation preview; Ho voice preview
- Our training: Mundari LoRA; Santali voice fine-tune vs a pre-set rule

Methodology and process for implementation

TEACHER INTERFACE

- Voice or typed input (Hindi)
- Lesson / activity / assessment modes
- Add a lesson from Hindi text
- Reading guide in Devanagari
- Android tablet app, or laptop browser
- Signed content sync, then no network

AI ENGINE — OFFLINE

- Hindi + Santali ASR — IndicConformer
- NMT — IndicTrans2, direct Indic ↔ Indic
- TTS — Piper, offline voice
- Teacher corrections reused first
- Doubtful output flagged (tier C)
- Long sentences streamed by clause

STUDENT OUTPUT

- Santali audio + Ol Chiki text
- Hindi response heard by teacher
- Worksheet, flashcards, lesson plan
- Green / yellow / red answer check
- Reading-fluency score (words/min)
- NIPUN Lakshya-tagged; NEP §4.11

• Analysis of the feasibility of the idea

- **On-device today:** Realme Pad Mini (4 GB) and a 2 GB Android 9 emulator, airplane mode: IndicConformer 120M, IndicTrans2 int8, Piper. New typed sentences translated on the tablet: p50 1.50 s (Realme), 0.60 s (emulator).
- **Published evidence:** Singh, Ekbal & Pakray (MMLoSo 2025): fine-tuned IndicTrans2 beats ByT5 for English ↔ Santali; we use the same family for Hindi ↔ Santali.
- **Benchmarks:** Hindi → Santali chrF++ 31.3 / 32.2 / 27.4 (IN22-Gen / IN22-Conv / FLORES-200), leakage-guarded; 300+ automated tests run in CI.
- **In progress, tested:** Mundari translation preview (our LoRA, held-out chrF++ 30.92); our fine-tuned Santali voice; Ho voice. Next: native-speaker review.

Offline · **Realme Pad Mini (4 GB)** · 2 GB emulator · laptop hub

0.70 s

lesson lines on the tablet, median (p90 1.20 s); 2 GB emulator 0.50 s

2.85 s

p90 hub, Hindi speech ≤ 17 words; Santali → Hindi p90 2.58 s

32.2 chrF++

Hindi → Santali, IN22-Conv public test set (n = 1,503)

24 / 24

app checks in airplane mode, on the Realme and the 2 GB emulator

Potential challenges / risks → Strategies for overcoming them

Ho & Mundari have far less parallel data than Santali (already built) → lower initial accuracy

Low-end tablets vary in RAM, CPU speed and heat — the weakest devices could exceed the 3-second latency limit

Processing children's voice data raises data-privacy concerns

Teachers with low digital literacy may resist adoption

LoRA on IndicTrans2 already gives a Mundari preview; add PALASH textbooks, community recordings and teacher corrections

INT8 models on the tablet, lessons pre-translated at sync, spoken lesson lines matched on-device, live latency readout

No cloud: speech stays on the tablet or school laptop; no child names in reports; reading-check audio deleted

Hindi, Santali and English interface, 18 pre-loaded NIPUN lessons — teachers start teaching, not configuring

IMPACT AND BENEFITS

• Potential impact on the target audience

DIRECT IMPACT & REACH

Built for PALASH's **1,041** MTB-MLE schools in 8 districts (JEPC) and the **5,000+** tribal-area primary schools named in the problem statement.



SCALABLE DEPLOYMENT PATH

Santali now; **Mundari** in preview; then **Ho**, a PALASH classroom pilot, all 24 Jharkhand districts, and Munda-language belts in Odisha, West Bengal and Bihar.



EDUCATIONAL FOUNDATION

Aims at the NIPUN Bharat goal: FLN by the end of Grade 3. Children learn to read and count in their home language first, then bridge to Hindi.



1,041

PALASH schools (JEPC, Jan 2026) — first deployment target

8 of 24

PALASH districts today → path to all 24

Rs. 0

Licence and cloud fees — open models, runs offline

18 lessons

every NIPUN Lakshya, Balvatika to Grade 3, with worksheets

Benefits of the solution (social, economic, operational, policy)

SOCIAL

Hindi-medium teachers can teach in the child's mother tongue — aims to lift comprehension before Grade 3

ECONOMIC

No licence fees or per-query cloud billing — IndicConformer and IndicTrans2 are MIT-licensed

OPERATIONAL

Offline on low-cost Android 9+ tablets, no new hiring; teacher corrections sync between tablets as signed files

POLICY

Supports NEP 2020 §4.11 (mother tongue), NIPUN Bharat FLN goals and SDG 4 (Quality Education)

